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Jordan Barber

Data Scientist

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Summary

Data Scientist with experience building and deploying cross-platform measurement and modelling products in cloud computing environments. I lead end-to-end client integrations, design scalable data pipelines in Databricks, and deliver production ML solutions used for campaign reporting and demographic reach estimation. PhD-trained in statistical modelling and large-scale computation.

Skills

Technical skills

Languages & Tools	Python (Pandas, NumPy, scikit-learn, PySpark), SQL, Databricks, Azure (Data Lake), MLflow, Git, high-performance computing (MPI, GPU acceleration)
ML & Modelling	Embedding models (BERT), optimisation-based modelling, supervised learning, Bayesian inference, model evaluation & metric design, Distributed processing (Spark), Monte Carlo methods

Experience

Data Scientist — **Apr 2025 — present**
Fifty5Blue (Formerly Kantar Media) — *London, UK*

- Led the end-to-end integration of a major global streaming client into a cross-media audience measurement product, coordinating across engineering and product teams to deliver a production-ready solution.
- Designed and built a semantic embedding-based methodology (BERT) to infer demographic distributions from impression-level data without labels, enabling scalable cross-platform reach measurement and non-naïve deduplication with Linear TV.
- Developed and productionised Databricks pipelines to compute cross-platform campaign reach across TV and multiple digital platforms. The workflow is parametrised and scheduled enabling the operations team to generate campaign reports and deliver daily live updates to client-facing dashboards.
- Enhanced a production ML modelling framework for demographic reach estimation by improving MLflow logging, strengthening model observability, and introducing new evaluation metrics.
- Integrated new digital platforms into the modelling pipeline, developing ingestion and preprocessing pipelines to support both single-platform and cross-media model training.

PhD Researcher – School of Physics and Astronomy — **Sept 2021 — Apr 2025**
Cardiff University — *Cardiff, UK*

- Designed and optimised large-scale simulation frameworks (C++/FORTRAN/Python), producing $\tilde{1}$0TB of data and running parallel workloads on HPC clusters leveraging Slurm, MPI, and GPU acceleration.
- Built statistical models to analyse high-dimensional simulation outputs, applying Bayesian inference and Monte Carlo sampling methods.
- Trained and deployed a machine learning model (normalising flows) to estimate posterior distributions from tens of millions of simulated data points.
- Developed new computational methods and improved legacy scientific codebases, improving performance and scalability.
- Supervised MSc dissertations and taught undergraduate programming and physics courses.

Research Scientist – Modelling observed supernovae — **Aug 2021 — Oct 2021**
Cardiff University — *Cardiff, UK*

- Developed an open-source Python tool using Gaussian processes and MCMC for parameter estimation on observational datasets.

Education

Doctor of Philosophy (PhD) in Physics & Astronomy, *Cardiff University* March 2025
Master of Physics (MPhys) in Astrophysics: 1st Class Honours (84%), *Cardiff University* July 2021

Publications

Barber, J., Chattopadhyay, D. and Antonini, F., 2024. *MNRAS*, 527(3), pp.7363-7381.

Barber, J. and Antonini, F., 2025. *MNRAS*, 538(2), pp.639–658.

Fantoccoli, F., Barber, J., et al, F., 2025. *MNRAS*, 538(1), pp.243–257.

Barber, J., 2025. PhD Thesis, Cardiff University.